PROJECT DOCUMENTATION:

TDMA Network Schedule Planner, Optimizer & EMANE Simulation Bridge

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1. Executive Summary & Problem Overview

Time Division Multiple Access (TDMA) enables shared wireless medium access by dividing time into discrete timeslots. In ad-hoc multi-hop wireless networks, static or uncontrolled transmission schedules cause severe packet loss due to two main types of channel interference:

1. **Direct Link Interference (Distance-1 / 1-Hop):** Adjacent nodes within radio range attempting to transmit at the same time.
2. **Hidden Terminal Interference (Distance-2 / 2-Hop):** Two non-adjacent nodes transmitting simultaneously to a shared intermediate neighbor, causing collisions at the receiver.

This project implements a centralized **Python Schedule Optimizer (The Brain)** to calculate a conflict-free Distance-2 TDMA schedule matrix while maximizing spatial reuse. It further integrates an **EMANE Integration Bridge (The Engine)** that converts the computed schedule into native XML event structures recognized by EMANE radio emulation environments.

2. Part 1: System Architecture & Design Process (The Brain)

2.1 Graph Representation & Distance Thresholding

- **Topology Modeling:** Built using Python's networkx library. Radios are represented as nodes $V = \{N_1, N_2, \dots, N_{16}\}$, and links are represented as undirected edges E.

- **Coverage Model:** Euclidean distance $d(N_i, N_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$ is computed between 2D node coordinates.
- **Threshold Rule:** An edge exists between N_i and N_j if and only if $d(N_i, N_j) \leq 500.0$ meters.

2.2 Distance-2 Conflict Detection Logic

To prevent both direct and hidden terminal collisions, every node constructs an extended conflict set $C(N_i)$:

- **1-Hop Neighbors (N^1):** Directly connected nodes within 500 meters.
- **2-Hop Neighbors (N^2):** Neighbors of 1-hop nodes.
- **Conflict Set Rule:** $C(N_i) = \left(N^1(N_i) \cup N^2(N_i)\right) \setminus \{N_i\}$.

2.3 Optimization Strategy & Spatial Reuse Heuristics

Finding the absolute minimum frame length for Distance-2 graph coloring is NP-hard. To minimize total frame length while enabling spatial reuse, a **Greedy Degree-Based Graph Coloring Heuristic** is implemented:

1. Calculate the conflict degree $|C(N_i)|$ for all nodes.
2. Sort nodes in descending order of conflict degree (most constrained nodes colored first).
3. Sequentially assign each node the lowest non-conflicting integer slot $s \in \{0,1,2, \dots\}$.
4. **Spatial Reuse Enforcement:** If two nodes N_a and N_b are separated by $>$ 2 hops, $N_b \notin C(N_a)$, allowing N_a and N_b to safely transmit in the exact same timeslot without interference.

3. Part 2: Physical Emulator Bridge Design (EMANE Setup)

3.1 Serialization Workflow

The integration bridge converts the discrete slot array generated in Part 1 into EMANE's native TDMA Radio Model XML schedule structure (emane_tdma_schedule.xml):

- **XML Schema:** Implements standard <emane-tdma-schedule> structure with <structure>, <multiframe>, <frame>, and <slot> elements.
- **Node Identifier Mapping:** Converts abstract node string keys (Node_01, Node_02, etc.) into numerical Network Emulation Module IDs (nem="1", nem="2", etc.).

3.2 Key EMANE Configuration Parameters

- **Frame Structure:** 1 frame per multiframe containing 9 timeslots.
- **Slot Duration:** 1000 μs (1 ms) per slot.
- **Radio Parameters:** Operating Frequency = 2.4 GHz (2,400,000,000 Hz), Bandwidth = 10 MHz, Data Rate = 1 Mbps, Guard Time = 100 μs.

4. Final Output & Result Values

4.1 Topology Configuration & Summary Values

- **Total Nodes Processed:** 16 Nodes (Grid arrangement from (0.0,0.0) to (900.0,900.0))
- **Configured Radio Range:** 500.0 Meters
- **Optimized Frame Length: 9 Unique Timeslots** (Slots 0 to 8)
- **Collision Status:** Verified Conflict-Free across all 1-hop and 2-hop links

4.2 Final Node-to-Slot Assignments

Node Identifier	Assigned Timeslot	Spatial Reuse Partners
Node_01	Slot 8	Node_04, Node_13, Node_16
Node_02	Slot 4	Node_14
Node_03	Slot 5	Node_15
Node_04	Slot 8	Node_01, Node_13, Node_16

Node Identifier	Assigned Timeslot	Spatial Reuse Partners
Node_05	Slot 6	Node_08
Node_06	Slot 0	Unique Slot (Central Node)
Node_07	Slot 1	Unique Slot
Node_08	Slot 6	Node_05
Node_09	Slot 7	Node_12
Node_10	Slot 2	Unique Slot
Node_11	Slot 3	Unique Slot
Node_12	Slot 7	Node_09
Node_13	Slot 8	Node_01, Node_04, Node_16
Node_14	Slot 4	Node_02
Node_15	Slot 5	Node_03
Node_16	Slot 8	Node_01, Node_04, Node_13

4.3 Structural TDMA Schedule Boolean Matrix (Slot x Node)

Plaintext

Slot \ Node | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16

Slot 00 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0

Slot 01 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0

Slot 02 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0

Slot 03 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0

Slot 04 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0

Slot 05 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0

Slot 06 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0

Slot 07 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0

Slot 08 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1

5. Verification & Conclusion

- **Conflict Avoidance:** Successfully eliminated direct link collisions and hidden terminal conflicts across all 16 node positions.
- **Spatial Reuse Efficiency:** Reused timeslots (e.g., Slot 8 assigned to 4 corner nodes separated by 900m) reduced frame length down to 9 slots.
- **EMANE Readiness:** Exported XML schema validates cleanly and can be loaded directly into an EMANE simulation daemon via emaneevent-tdmaschedule.